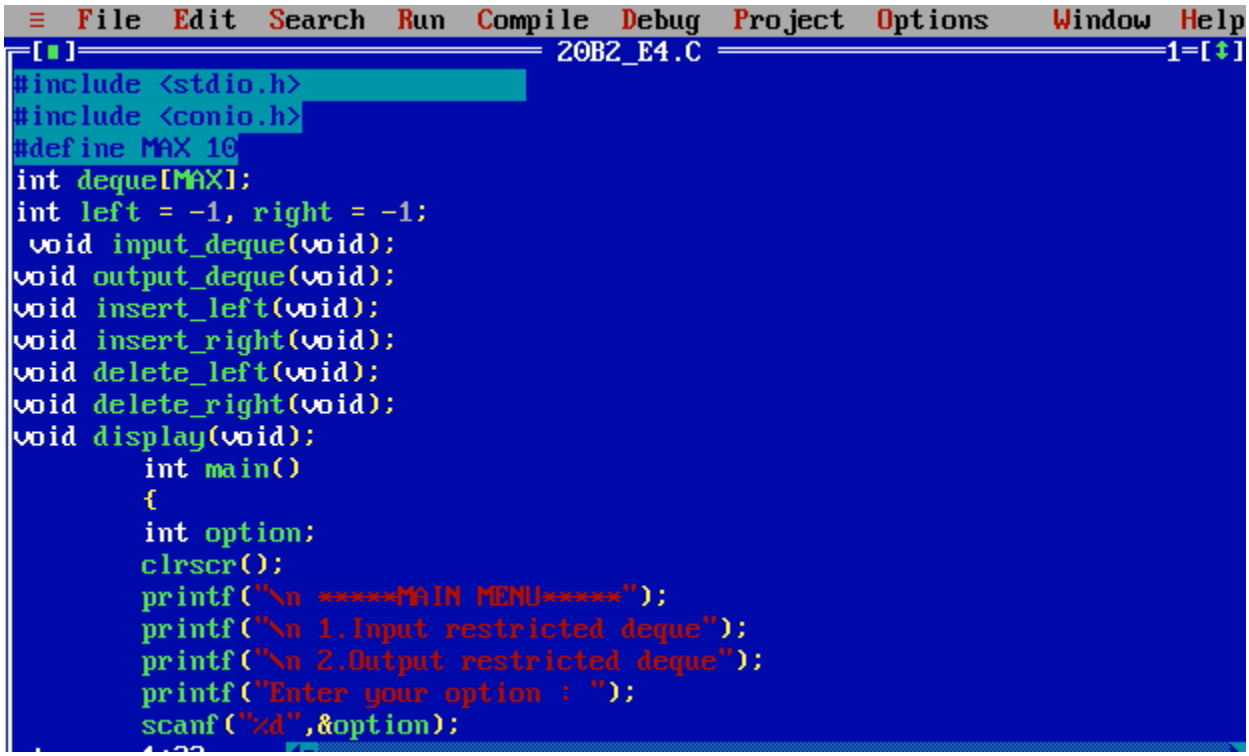
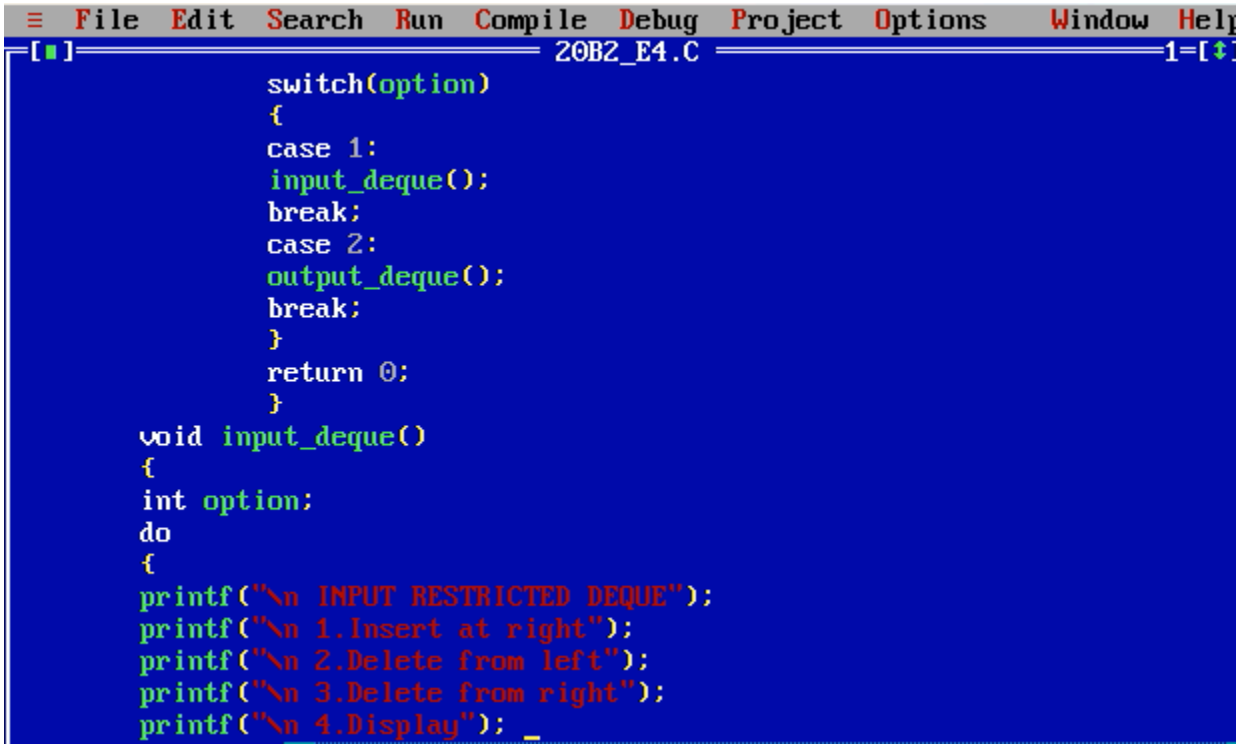
Name:Pithadia Yug Shailesh Rollno: B\20

ExpNo4: Dequeue





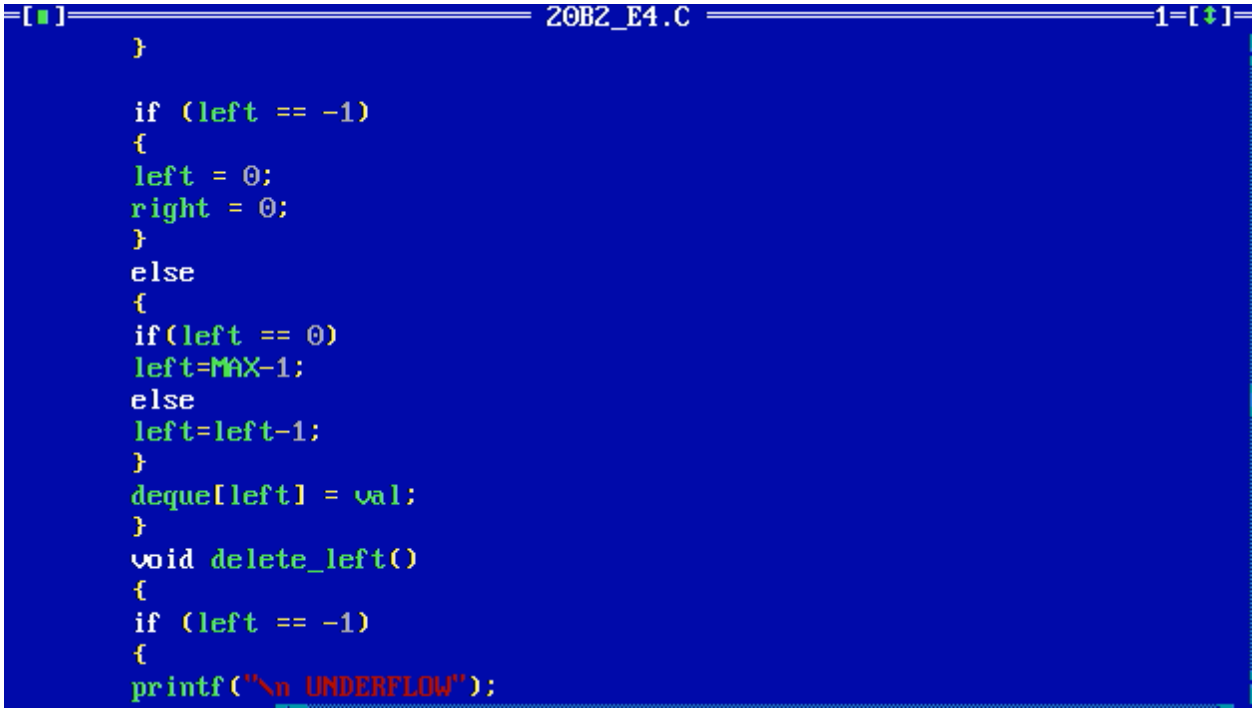
```
=[■]═══════════ 20B2_E4.C ═══════════1=[↕]=
        printf("\n 5.Quit");
        printf("\n Enter your option : ");
        scanf("%d",&option);
                switch(option)
                {
                case 1:
                insert_right();
                break;
                case 2:
                delete_left();
                break;
                case 3:
                delete_right();
                break;
                case 4:
                display();
                break;
                }
                        }while(option!=5);

                }
```

```
 ≡  File  Edit  Search  Run  Compile  Debug  Project  Options  Window  Help
=[■]═══════════ 20B2_E4.C ═══════════1=[↕]=
        void output_deque()
        {
        int option;
        do
        {
        printf("OUTPUT RESTRICTED DEQUE");
        printf("\n 1.Insert at right");
        printf("\n 2.Insert at left");
        printf("\n 3.Delete from left");
        printf("\n 4.Display");
        printf("\n 5.Quit");
        printf("\n Enter your option : ");
        scanf("%d",&option);
        switch(option)
        {
        case 1:
        insert_right();
        break;
        case 2:
        insert_left();
        break;
```

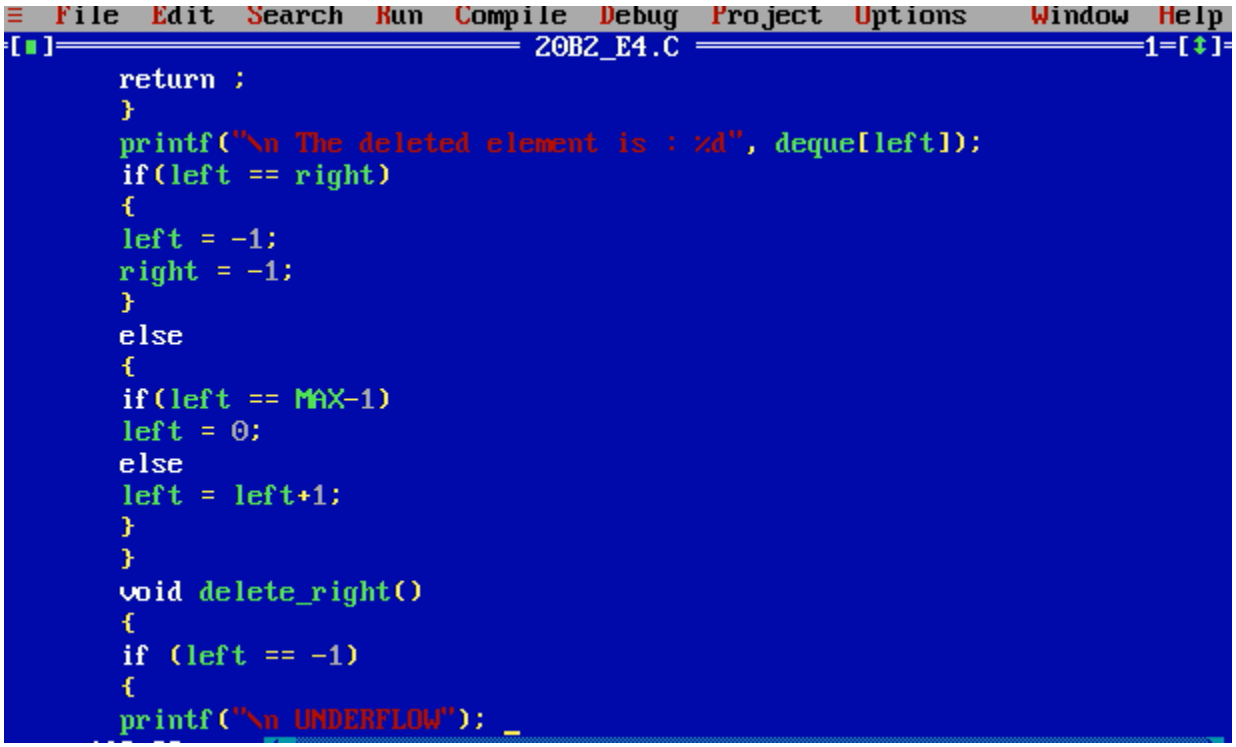
```
File  Edit  Search  Run  Compile  Debug  Project  Options    Window  Help
=[■]═══════════════════════ 20B2_E4.C ═══════════════════════1=[↕]
        case 3:
        delete_left();
        break;
        case 4:
        display();
        break;
        }
                }while(option!=5);
        }
        void insert_right()
        {
        int val;
        printf("\n Enter the value to be added:");
        scanf("%d", &val);
        if((left == 0 && right == MAX-1) || (left == right+1))
        {
        printf("\n OVERFLOW");
        return;
        }
        if (left == -1)
        {
```

```
File  Edit  Search  Run  Compile  Debug  Project  Options    Window  Help
=[■]═══════════════════════ 20B2_E4.C ═══════════════════════1=[↕]
        left = 0;
        right = 0;
        }
        else
        {
        if(right == MAX-1)
        right = 0;
        else
        right = right+1;
        }
        deque[right] = val ;
        }
        void insert_left()
        {
        int val;
        printf("\n Enter the value to be added:");
        scanf("%d", &val);
        if((left == 0 && right == MAX-1) || (left == right+1))
        {
        printf("\n Overflow");
        return;
```

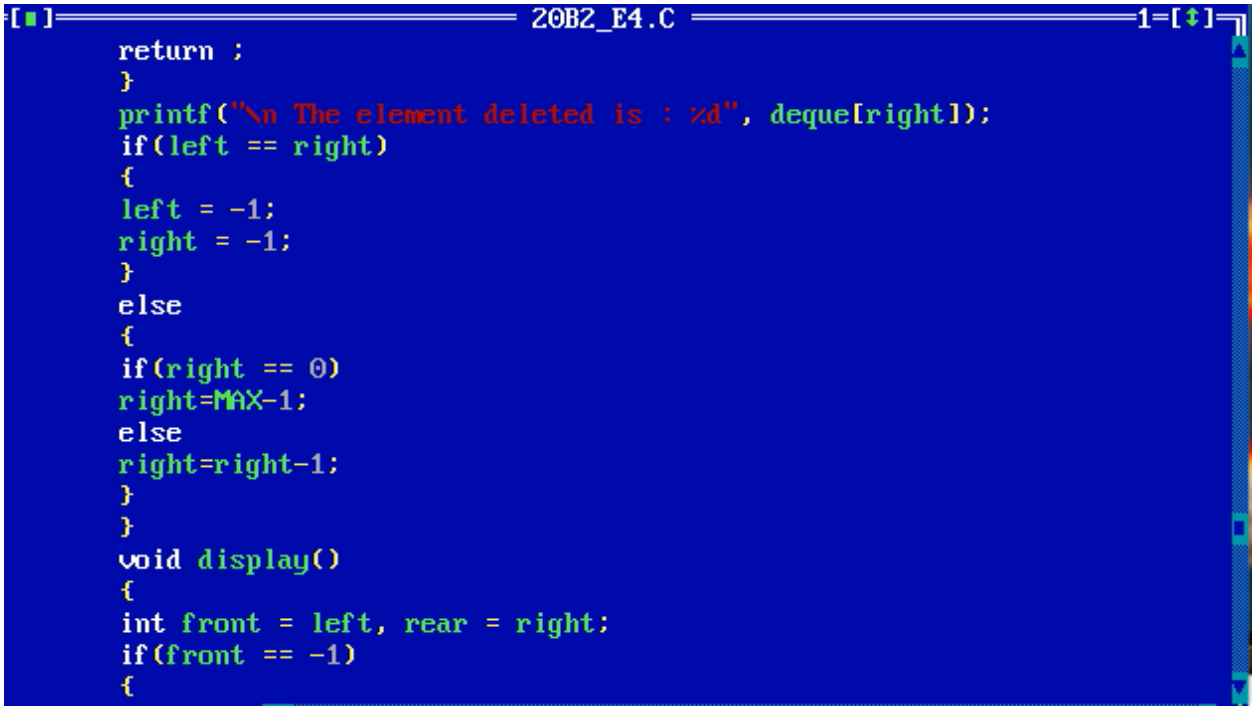


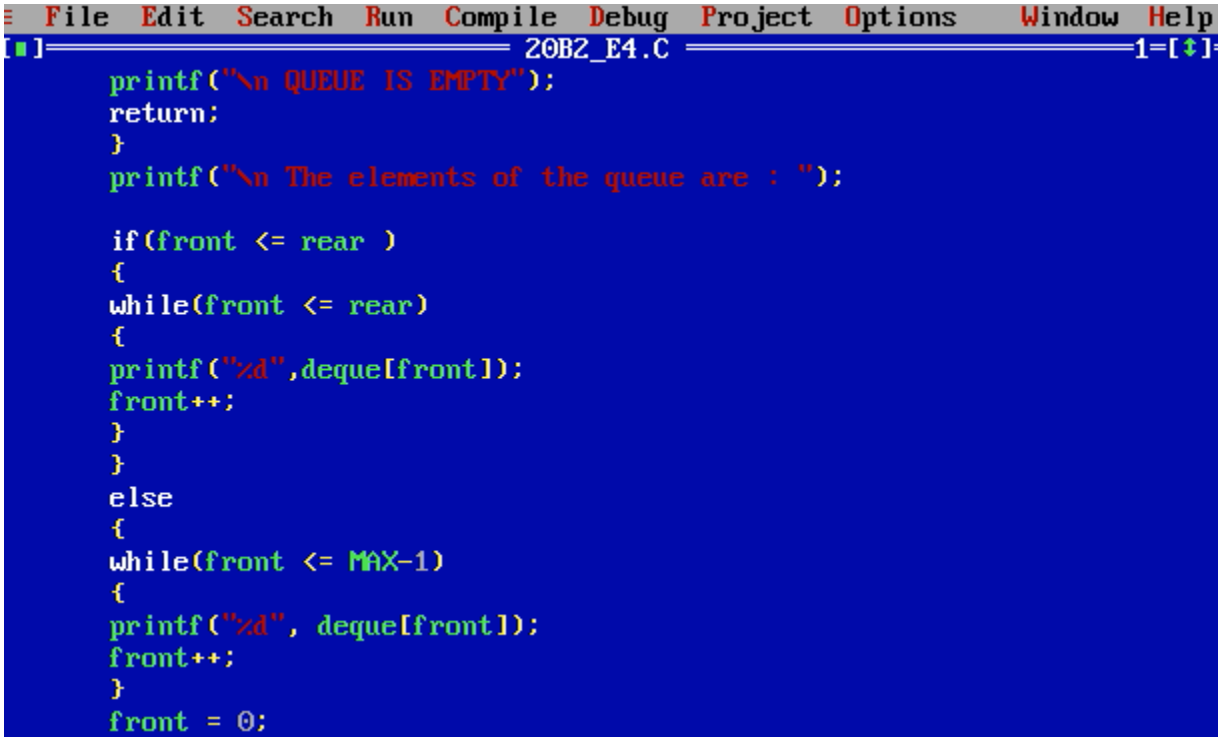
```
[■]═══════════ 20B2_E4.C ═══════════1=[↕]
        }

        if (left == -1)
        {
        left = 0;
        right = 0;
        }
        else
        {
        if(left == 0)
        left=MAX-1;
        else
        left=left-1;
        }
        deque[left] = val;
        }
        void delete_left()
        {
        if (left == -1)
        {
        printf("\n UNDERFLOW");
```

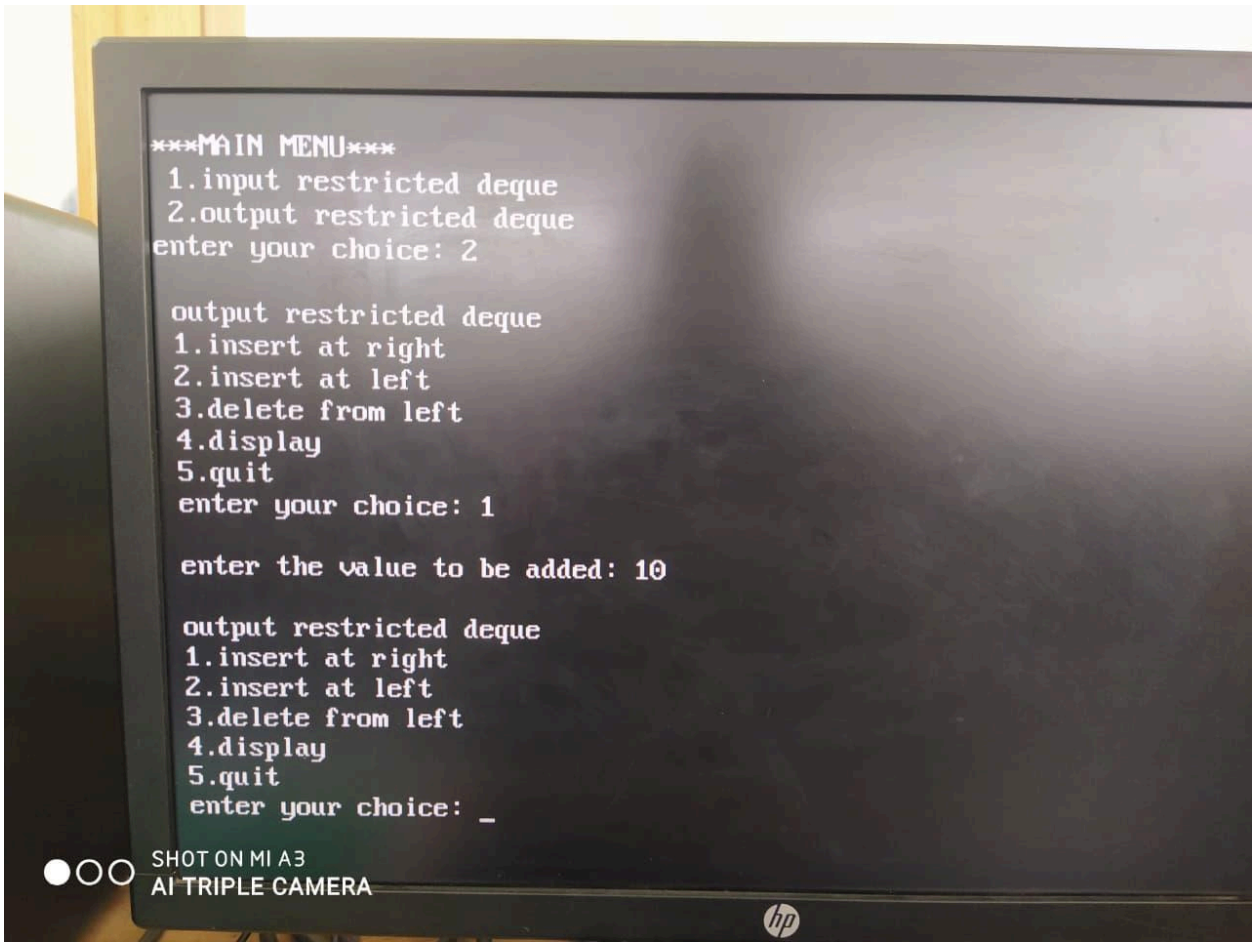


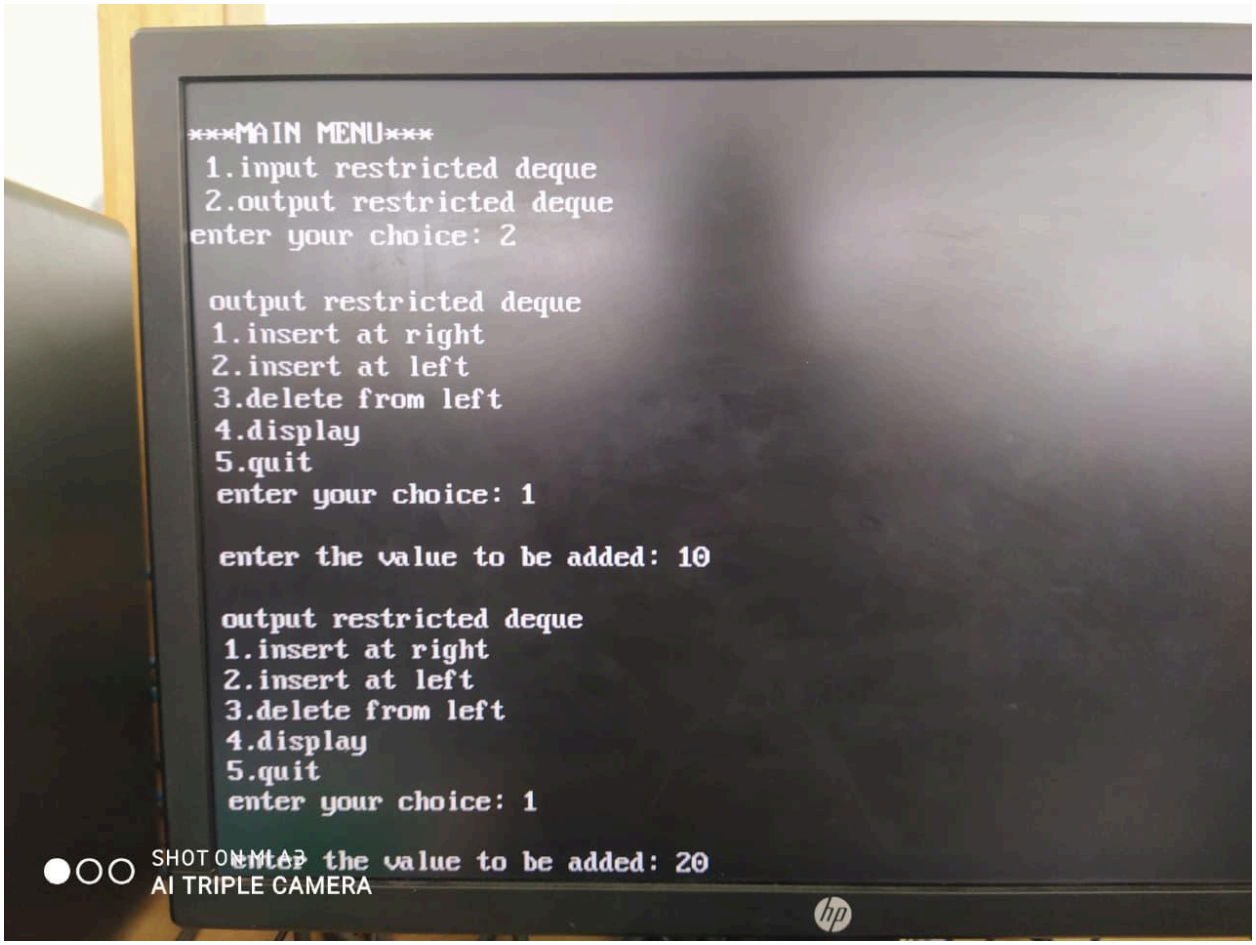
```
≡ File  Edit  Search  Run  Compile  Debug  Project  Uptions   Window  Help
[■]═══════════ 20B2_E4.C ═══════════1=[↕]
       return ;
       }
       printf("\n The deleted element is : %d", deque[left]);
       if(left == right)
       {
       left = -1;
       right = -1;
       }
       else
       {
       if(left == MAX-1)
       left = 0;
       else
       left = left+1;
       }
       }
       void delete_right()
       {
       if (left == -1)
       {
       printf("\n UNDERFLOW");
```

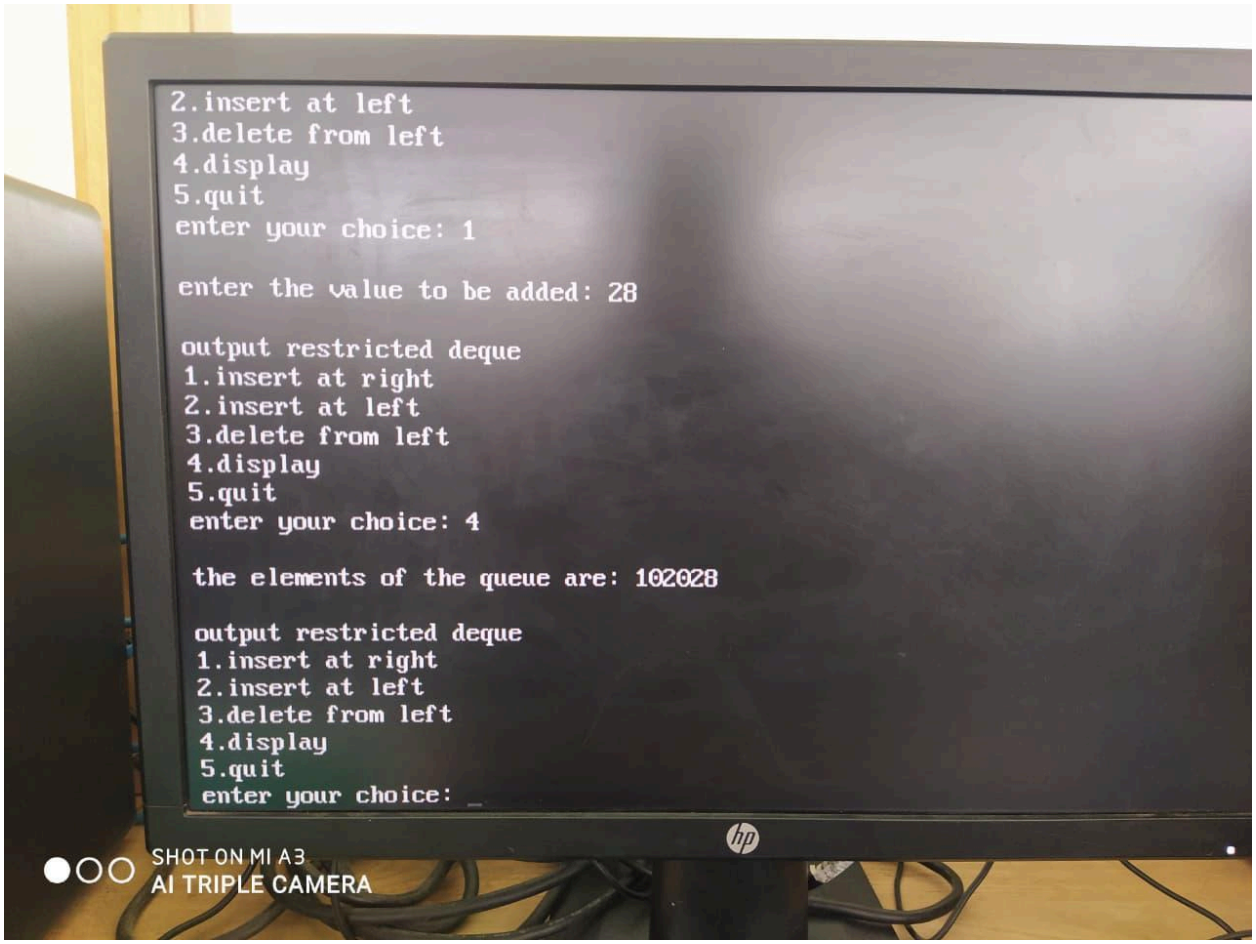

20B2_E4.C
return ;
}
printf("\n The element deleted is : %d", deque[right]);
if(left == right)
{
left = -1;
right = -1;
}
else
{
if(right == 0)
right=MAX-1;
else
right=right-1;
}
}
void display()
{
int front = left, rear = right;
if(front == -1)
{

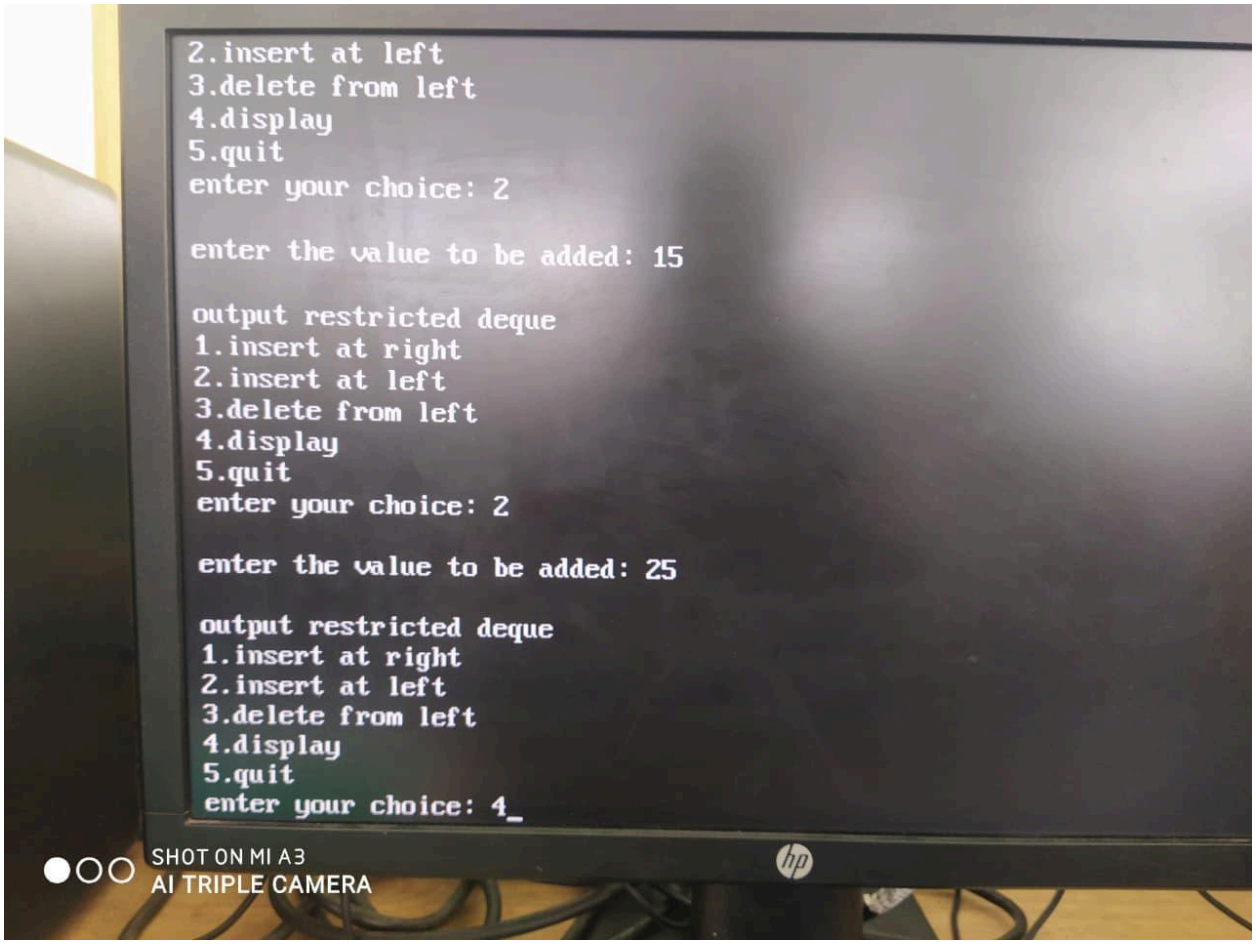

File Edit Search Run Compile Debug Project Options Window Help
20B2_E4.C
printf("\n QUEUE IS EMPTY");
return;
}
printf("\n The elements of the queue are : ");
if(front <= rear)
{
while(front <= rear)
{
printf("%d",deque[front]);
front++;
}
}
else
{
while(front <= MAX-1)
{
printf("%d", deque[front]);
front++;
}
front = 0;

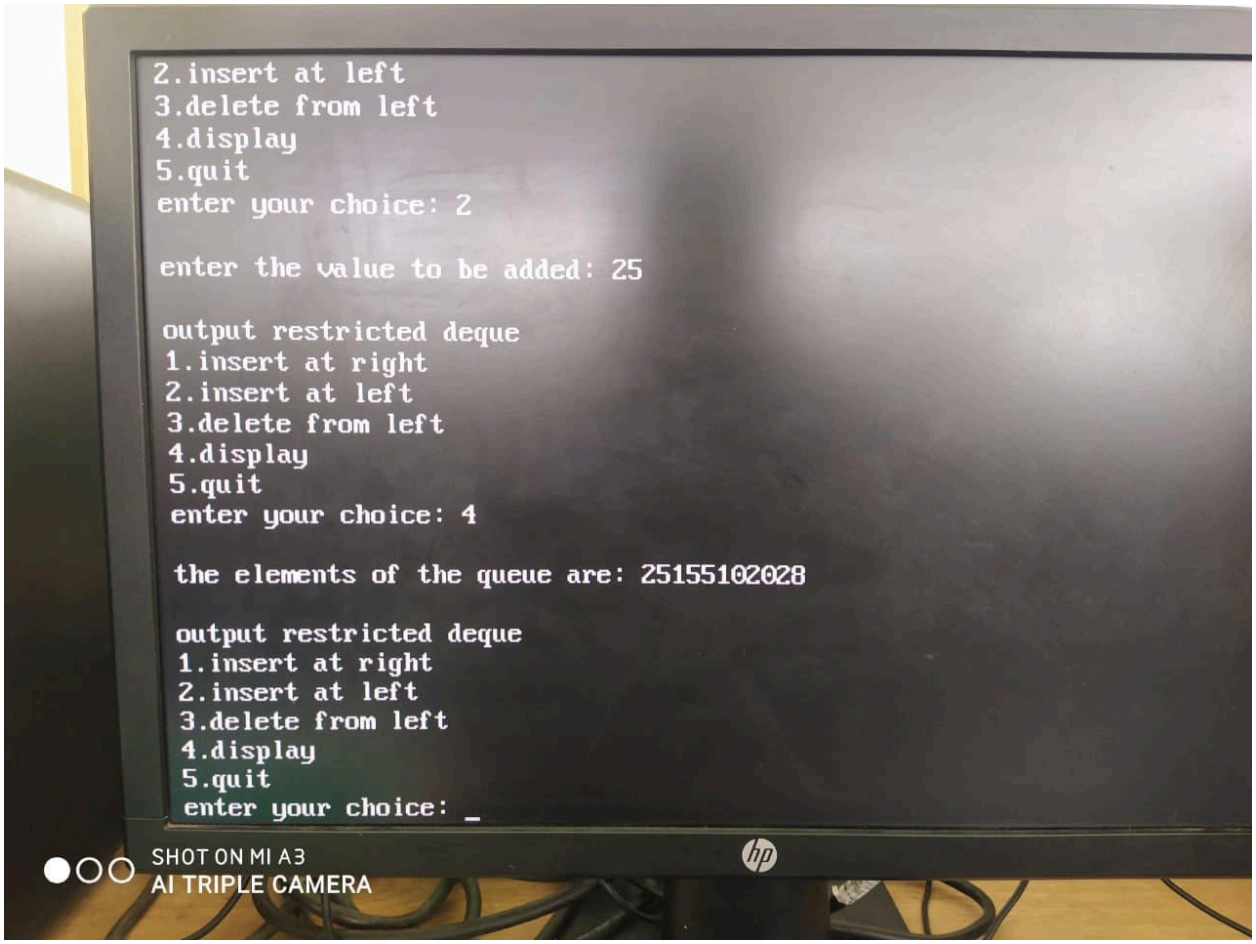
```
        {
        printf("%d",deque[front]);
        front++;
        }
        }
        else
        {
        while(front <= MAX-1)
        {
        printf("%d", deque[front]);
        front++;
        }
        front = 0;
        while(front <= rear)
        {
        printf("%d",deque[front]);
        front++;
        }
        }
        printf("\n");
      }
```

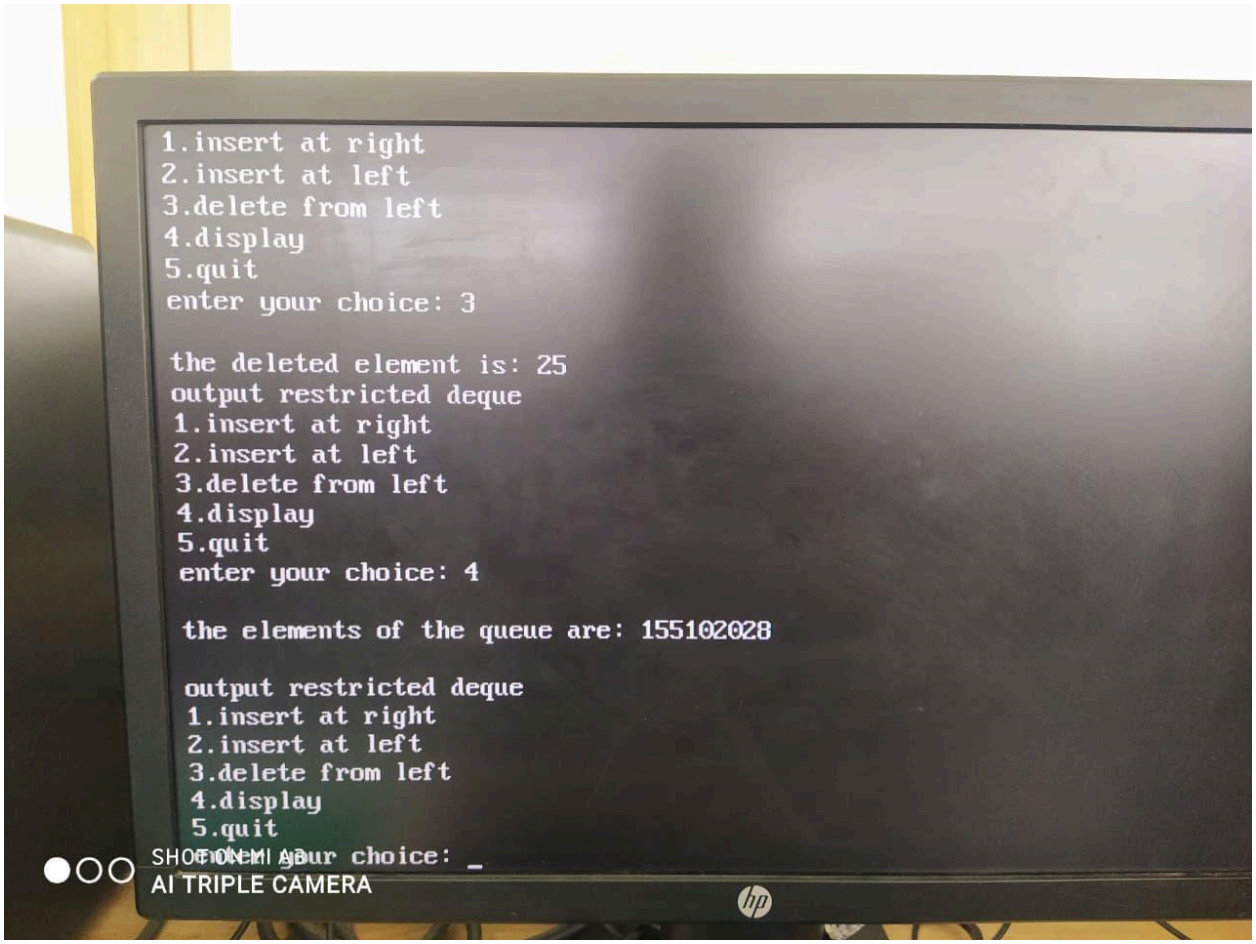



MAIN MENU
1.input restricted deque
2.output restricted deque
enter your choice: 2
output restricted deque
1.insert at right
2.insert at left
3.delete from left
4.display
5.quit
enter your choice: 1
enter the value to be added: 10
output restricted deque
1.insert at right
2.insert at left
3.delete from left
4.display
5.quit
enter your choice: 1
enter the value to be added: 20
SHOT ON MI A3
AI TRIPLE CAMERA


2.insert at left
3.delete from left
4.display
5.quit
enter your choice: 1
enter the value to be added: 28
output restricted deque
1.insert at right
2.insert at left
3.delete from left
4.display
5.quit
enter your choice: 4
the elements of the queue are: 102028
output restricted deque
1.insert at right
2.insert at left
3.delete from left
4.display
5.quit
enter your choice: _
SHOT ON MI A3
AI TRIPLE CAMERA


2.insert at left
3.delete from left
4.display
5.quit
enter your choice: 2
enter the value to be added: 15
output restricted deque
1.insert at right
2.insert at left
3.delete from left
4.display
5.quit
enter your choice: 2
enter the value to be added: 25
output restricted deque
1.insert at right
2.insert at left
3.delete from left
4.display
5.quit
enter your choice: 4
SHOT ON MI A3
AI TRIPLE CAMERA


2.insert at left
3.delete from left
4.display
5.quit
enter your choice: 2
enter the value to be added: 25
output restricted deque
1.insert at right
2.insert at left
3.delete from left
4.display
5.quit
enter your choice: 4
the elements of the queue are: 25155102028
output restricted deque
1.insert at right
2.insert at left
3.delete from left
4.display
5.quit
enter your choice: _
SHOT ON MI A3
AI TRIPLE CAMERA


1.insert at right
2.insert at left
3.delete from left
4.display
5.quit
enter your choice: 3
the deleted element is: 25
output restricted deque
1.insert at right
2.insert at left
3.delete from left
4.display
5.quit
enter your choice: 4
the elements of the queue are: 155102028
output restricted deque
1.insert at right
2.insert at left
3.delete from left
4.display
5.quit
enter your choice:
AI TRIPLE CAMERA